NIKHIL MAHADEVAN

Nikhil.Mahadevan@uga.edu • linkedin.com/in/nmahadevan12 • github.com/nmahadevan12 • nmahadevan12.github.io

Education

University of Georgia | Athens, GA **Aug 2024 - May 2028**
B.S. in Electrical Engineering, B.S. in Computer Engineering **GPA: 3.93**
Relevant Coursework: Embedded Systems, Sensors and Transducers, Linear Systems, Electronics, Logic Design, Discrete Math
Honors & Memberships: Eagle Scout**,** UGA Dean's List, UGA IEEE MicroMouse, UGA Outdoor Club

Experience

Flight Software Engineer | University of Georgia Small Satellite Research Laboratory **Sep 2026 - Present**
- Developing CubeSat flight software integrating C++ active components in NASA F Prime with low-level C device drivers under Zephyr RTOS on an STM32 MCU
- Implemented an on-board computer 7-state mode controller in C++ featuring 10-tick voltage debounce, request priority arbitration, and fault-tolerant safe shutdown sequencing
- Refactored legacy file-recycler service, eliminating build failures by replacing raw strings with path abstractions, resolving symlink traversal vulnerabilities, and enforcing thread-safe mutex synchronization
- Implementing UART and RS-485 serial drivers in C via the Zephyr device driver model to support multi-node telemetry routing, validating driver integration in Renode virtual environments

Embedded Systems Research Intern | University of Georgia Vertically Integrated Projects for Research **Aug 2026 - Present**
- Developing C++ drivers across I2C and SPI interfaces to acquire real-time IMU, ambient temperature, and relative humidity telemetry on an ATmega328P-based canine health monitor
- Implementing serial communication via an HC-05 Bluetooth module using hardware UART to stream structured multi-sensor data packets to a host application in real time

Electronics Lab Assistant & Engineering Peer Tutor| University of Georgia **Sep 2025 - Present**
- Instructing 100+ students across two labs on analog filters, oscillators, transformers and BJT amplifiers with SPICE verification
- Guiding students with TTL circuit builds and benchtop debugging using oscilloscopes, logic analyzers and multimeters
- 4Tutored 35+ students across courses mainly focused on Logic Design (CSEE 2220) and Circuit Analysis (ECSE 2170)

Projects

Edge AI Wake-Word Engine **Aug 2026**
- Engineered register-level C firmware on an STM32L4 (ARM Cortex-M4) for real-time PDM audio acquisition
- Implemented a DC-blocking filter and energy detector in C to isolate speech buffers and reject background noise
- Developed a 921.6 kbps UART streaming driver with SysTick timestamping to frame and transfer raw audio buffers to a host PC
- Deployed an on-device MFCC-based logistic regression model on the Cortex-M4 (< 2 KB RAM footprint, sub-millisecond score latency), achieving 92% validation accuracy on a 600-sample balanced dataset

2-Axis Light Tracker (ELEE 4230) **Jun 2026 - Jul 2026**
- Built a battery-powered dual-axis light tracker in bare-metal C on an ATmega328P using MOSFET H-bridges to drive motors
- Integrated an LDR array with Hall-effect and potentiometer feedback for tilt limit protection
- Implemented a fixed-point Kalman filter to smooth ADC readings, reducing peak noise from 3 LSBs to < 1 LSB without floating-point overhead
- Engineered a register-level 115.2 kbps UART driver to serialize and stream real-time Kalman-filtered tracking telemetry
- Designed a 4-layer PCB in KiCad with dedicated ground planes to isolate motor switching noise from sensitive analog sensor traces

Multifunction Instrumentation System (ECSE 2920) **Jan 2026 - Apr 2026**
- Built a Raspberry Pi DMM and function generator in Python powered by a ±12V split-rail supply with active virtual ground
- Designed a discrete SAR ADC via SPI digipot and comparator for a 0–5V voltmeter, 500Ω-10 kΩ ohmmeter, and ±5V reference
- Built function generators (0–10Vpk, 0–10 kHz) and a 0–10 kHz frequency counter buffered by op-amp conditioning stages
- Improved DC resolution 6x and voltmeter precision 2x via digipot calibration in software
- Designed a 4-layer mixed-signal Pi HAT PCB with dedicated power and ground planes

Technical Skills

Languages: C, C++, ARM Assembly, Python, Java, MATLAB
Embedded Systems: ARM Cortex-M4 (STM32), ARM Cortex-A53 (BCM2837), ATmega328P, ESP32, Zephyr RTOS, NASA F Prime
Hardware & EDA: Altium Designer, KiCad, 4-Layer PCB Design, SPICE, Autodesk Fusion
Tools & Test Equipment: Renode, Git, Oscilloscopes, Logic Analyzers, Multimeters, Soldering